



ExSEnt: Extrema-Segmented Entropy analysis of time series

Sara Kamali¹ · Fabiano Baroni¹ · Pablo Varona¹

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Abstract

We introduce Extrema-Segmented Entropy (ExSEnt), a feature-decomposed framework for quantifying time-series complexity that separates temporal from amplitude contributions. The method partitions a signal into monotonic segments by detecting sign changes in the first-order increments. For each segment, it extracts the interval duration and the net amplitude change, generating two sequences that reflect timing and magnitude variability, respectively. Complexity is then quantified by computing sample entropy on durations and amplitudes, together with their joint entropy. This decomposition reveals whether overall variability is driven by duration, amplitude, or their coupling, providing a richer and more interpretable characterization than unidimensional metrics. We validate ExSEnt on canonical nonlinear dynamical systems (Logistic map, Rössler system, Rulkov map), demonstrating its ability to track variability changes across control parameter sweeps and detect transitions between regular to chaotic regimes. Then, we illustrate the empirical utility of ExSEnt metrics to isolate feature-specific components of complexity in real data (electromyography and ankle acceleration in Parkinson's disease). Thus, ExSEnt complements existing entropy measures by attributing complexity to distinct signal features, improving interpretability and supporting applications in a broad range of domains, including physiology, finance, and geoscience.

Keywords Time-series complexity · Sample entropy · Extrema-based segmentation · Event-based segmentation · Information theory · Bifurcation analysis

1 Introduction

The study of time-series complexity, in the framework of structural organization, is central to understanding the behavior and predictability of nonlinear dynamical systems. Many natural and engineered processes—from physiological rhythms to markets and climate—exhibit intricate temporal organization that demands robust quantitative descriptors [1–5]. Entropy-based metrics are widely used to quantify uncertainty and diversity of observed patterns, and (when defined as entropy rates or conditional entropies) to quantify aspects of predictability in temporal data [6–13], with extensive applications in physiological signal analysis [14–

18], finance [19, 20], climate [21], and network dynamics [22, 23]. Foundational work links macroscopic complexity to healthy adaptive function, with characteristic loss of complexity in aging and disease [24, 25], motivating precise and interpretable complexity quantification.

The concept of "complexity" itself has been defined differently across disciplines, including algorithmic complexity [26], dynamical and chaotic complexity [27], statistical complexity [28], complexity as interplay of segregation and integration [29], and entropy-based irregularity measures for physiological signals [9, 14, 30]. In the present work, we propose a complexity measure as characterization of the variability and temporal organization of extrema-derived amplitude and duration patterns.

Classical entropy measures probe different aspects of variability and predictability (from marginal symbol diversity to sequence-dependent regularity and entropy-rate concepts), depending on whether temporal ordering is incorporated in the definition. Shannon entropy quantifies the uncertainty of a probability distribution over symbols (or discretized values) [6]. Many "symbolic" entropies are computed by applying Shannon's formula to the empirical distribution of patterns

✉ Sara Kamali
sara.kamali@uam.es

Fabiano Baroni
fabiano.baroni@uam.es

Pablo Varona
pablo.varona@uam.es

¹ GNB. Dpto. de Ingenieria Informatica., Autonomous University of Madrid, C. Francisco Tomás y Valiente, 11, 28049 Madrid, Spain

extracted from the time series. Permutation entropy (PE) is Shannon entropy of ordinal patterns, i.e., the rank order of samples within an embedding vector [2, 31]. In its basic form, PE discards absolute amplitude information; several variants explicitly reincorporate amplitude or fluctuation content, including Weighted-PE (WPE) [10] and Amplitude-Aware PE (AAPE) [32]. WPE weights each ordinal pattern occurrence by a local amplitude statistic (e.g., variance/energy) of the samples forming the pattern, whereas AAPE incorporates relative magnitudes directly into the symbolization step, embedding amplitude information into the pattern definition.

Related symbolic approaches define patterns from local changes rather than ranks. Increment entropy and its multiscale extension quantify the distribution of increment-coded patterns [33, 34]. Slope entropy encodes subsequences by thresholding successive differences (positive, negative, or near-zero), thereby incorporating both sign and (thresholded) magnitude of local changes [35]. Dispersion entropy and its multiscale/refined variants quantify the distribution of dispersion-class patterns and are designed for stability and computational efficiency in biomedical data [36, 37]. Practical guidance on parameterization and pitfalls for the PE family is well documented [31, 38–41].

Beyond symbolic entropies, Kolmogorov–Sinai entropy formalizes a dynamical entropy rate for chaotic systems [42, 43]. Approximate entropy (ApEn) and sample entropy (SampEn) are similarity-based regularity statistics computed from counts of similar templates in reconstructed state space; they are commonly interpreted as estimating a conditional probability of pattern continuation [44, 45]. Multiscale entropy (MSE) extends such regularity estimation by evaluating entropy after coarse-graining the signal across scales, with refinements such as refined composite MSE improving robustness for short/noisy recordings [9, 17, 46].

Despite broad uptake, a key limitation persists: conventional entropy measures rarely disentangle distinct components of complexity within a single signal. In real systems, variability arises from: (i) variability in the timing of upward/downward drifts (durations of such drifts), and (ii) variability in the magnitudes of those drifts (amplitude changes). Standard estimators typically conflate these contributions. For example, ordinal analyses emphasize rank order (timing/order relations) but not magnitudes unless explicitly modified (AAPE, WPE) [2, 10, 32, 47], and many MSE implementations inherit this ambiguity [9, 46, 48, 49]. Although multivariate and bivariate entropies and mutual-information frameworks capture dependencies across channels or features [50–56], they do not provide a feature-level decomposition of a single time series into temporal- vs. amplitude-driven components of complexity. The need for such decomposition is underscored in contexts where timing (phase/duration) governs behavior more than amplitude (e.g., specific oscillatory neural dynamics): signal-processing the-

ory formally separates instantaneous phase and amplitude via analytic signals and the Hilbert transform [57, 58], but an entropy formulation that quantifies their independent contributions has been missing.

Such decomposition provides insight and understanding in multiple contexts. For example, in neural time series, temporal variations relate to oscillatory dynamics, synchrony, and sequential/phase relations; amplitude variations reflect power/gain changes and can be disproportionately affected by artifacts or non-neural influences, particularly when identifying functional variability. Conflation reduces the interpretability of EEG complexity biomarkers. Studies comparing entropy indices in anesthesia and clinical EEG highlight sensitivity to algorithmic choices and to different noise/artifact regimes [59–64], and emphasize the importance of scale-aware, robust, and interpretable metrics [40, 48, 49]. A decomposition that attributes overall complexity to temporal versus amplitude-driven irregularities could sharpen inference, improve predictive performance, and clarify mechanistic hypotheses.

To address these limitations, we introduce Extrema-Segmented Entropy (ExSEnt), a feature-driven, noise-robust framework that segments a signal by the sign of its increments, forming monotone segments between extrema. For each segment, we extract duration $\mathcal{D}$ and net amplitude change $\mathcal{A}$, and then compute SampEn on the $\mathcal{D}$ and $\mathcal{A}$ sequences separately ($\mathcal{H}_{\mathcal{D}}$, $\mathcal{H}_{\mathcal{A}}$) alongside their joint entropy ($\mathcal{H}_{\mathcal{DA}}$). This yields an interpretable decomposition, revealing whether complexity arises primarily from timing variability, amplitude variability, or their interaction. If $\mathcal{H}_{\mathcal{DA}} < \min(\mathcal{H}_{\mathcal{D}}, \mathcal{H}_{\mathcal{A}})$, there is some level of coherency in the variations of $\mathcal{D}$ and $\mathcal{A}$, which makes their paired sequence more predictable than when considered alone; if $\mathcal{H}_{\mathcal{DA}} > \max(\mathcal{H}_{\mathcal{D}}, \mathcal{H}_{\mathcal{A}})$, the features contribute independently to the complexity of the signal with lower predictability of their paired sequence.

We validate ExSEnt on canonical chaotic systems—the Logistic map, the Rössler system, and the Rulkov map [65–67]—which undergo well-characterized transitions between periodic and chaotic regimes. Unlike single-number metrics, ExSEnt tracks $\mathcal{H}_{\mathcal{D}}$, $\mathcal{H}_{\mathcal{A}}$, and $\mathcal{H}_{\mathcal{DA}}$ along control parameter sweeps and exposes whether complexity changes are driven by timing variability, amplitude variability, or their coupling, thereby enhancing interpretability. Beyond nonlinear systems analysis, this feature-level decomposition is a plausible candidate for EEG/MEG biomarkers in epilepsy, anesthesia, sleep, and disorders of consciousness, and for biometric and forensic signal analysis, where the component of complexity affects robustness and specificity. To illustrate these benefits in real-world biosignals, we evaluated complexity in electromyography (EMG) at rest versus movement, and in lower limb motion acceleration of individuals with Parkin-

son's disease (PD), to demonstrate how ExSEnt disentangles temporal- versus amplitude-driven contributions.

The remainder is organized as follows. Section 2 details ExSEnt's formulation and implementation. Section 3 applies ExSEnt to chaotic benchmarks and real physiological data, demonstrating sensitivity to regime transitions and the ability to attribute complexity changes to timing and/or amplitude. Sections 4 and 5 discuss the observations, broader implications, and applications.

2 Methodology

We propose a novel method, *ExSEnt*, to quantify the complexity of a signal by leveraging its local bivariate variations. The core idea is to segment the signal based on local extrema, extract duration and amplitude features from each segment, and then apply entropy measures to evaluate their complexity. The methodology consists of three steps:

Segmentation based on local extrema: The local minima and maxima of the signal are extracted. Every interval between two consecutive local extrema is considered as a segment.

Feature extraction: Two key features, i.e., the duration and the net amplitude change, are extracted for each segment. Figure 1 depicts this step for one segment.

Entropy computation: SampEn is computed on the extracted features and their paired sequence.

Each of these steps is described in detail in the following subsections.

2.1 Local extrema-based segmentation

Let $X(t) = x(t_1), x(t_2), \dots, x(t_N)$ denote the time series of interest. To detect the local extrema, we define the first-order amplitude increment for the k^{th} sample as:

$$\Delta x_k = x(t_k) - x(t_{k-1}). \quad (1)$$

A segment is defined as a continuous set of data points, during which the sign of all Δx_k values remains constant. Hence, a local extremum, $\mathcal{E}_i$, is defined as a point where the sign of Δx_k differs from that of the subsequent amplitude increment, Δx_{k+1} . To enhance robustness against noise, a noise threshold θ is applied, set to λ times the interquartile range (IQR) of the set of all increments of the time series, i.e., $\{\Delta x_k\}$:

$$\theta = \lambda \times \text{IQR}(\{\Delta x_k\}). \quad (2)$$

Hence, an increment Δx_{k+1} marks the beginning of a new segment only if $|\Delta x_{k+1}| > \theta$ and $\Delta x_k \Delta x_{k+1} < 0$.

Thus, the noise tolerance depends on λ . A small λ captures finer fluctuations, whereas a larger value suppresses

noise-induced variations. So, this parameter may be tuned depending on the dataset and noise characteristics.

2.2 Feature extraction

After segmentation, we extract two fundamental features per segment: first, the segment duration ($\mathcal{D}$), i.e., the interval between the two extreme points, $\mathcal{E}_i$ and $\mathcal{E}_{i-1}$:

$$\mathcal{D}_i = t_{\mathcal{E}_i} - t_{\mathcal{E}_{i-1}} \quad (3)$$

second, the amplitude ($\mathcal{A}$) of the segment, i.e., the net amplitude change between the consecutive local extrema:

$$\mathcal{A}_i = x(t_{\mathcal{E}_i}) - x(t_{\mathcal{E}_{i-1}}), \quad (4)$$

where the sign of the magnitude differences is preserved.

These features form two separate time series, $\{\mathcal{D}_i\}$ and $\{\mathcal{A}_i\}$, which are the basic elements for computing the ExSEnt metrics. As illustrated in Fig. 1, our goal is to study the pattern of local extrema-based segment duration and amplitude by analyzing the sequence of extracted $\{\mathcal{D}_i\}$ and $\{\mathcal{A}_i\}$ for $i = 1, \dots, M$.

By segmenting the time series based on the monotonic parts between its extrema, the signal is summarized into extrema-based segments characterized in terms of duration and magnitude. Hence, the duration and amplitude of consistent trends are preserved, while the exact morphology of the waveform between extrema is discarded.

2.3 Sample entropy characterization

We selected SampEn to quantify the variability of the extracted features because it is particularly advantageous for small datasets, making it suitable for real-world applications. It measures the probability that two sequences of length m , similar within a tolerance r , remain similar when extended to length $m + 1$. The mathematical definition is:

$$\begin{aligned} \text{SampEn}(m, r, N) &= -\log \left(\frac{\sum_{i=1}^{N-m-1} A_i(m, r)}{\sum_{i=1}^{N-m} B_i(m, r)} \right) \\ &= -\log \left(\frac{A}{B} \right), \end{aligned} \quad (5)$$

where m is the embedding dimension (length) of the sequences and r is the tolerance threshold to measure similarity between two sequences, which is typically set as a fraction of the Standard Deviation (SD) of the time series. N is the total number of data points. The SampEn computation is based on the frequency of appearance of similar (matching) templates. A template of length m , $\mathbf{T}^{(m)}$, refers to m consecutive data points in the series,

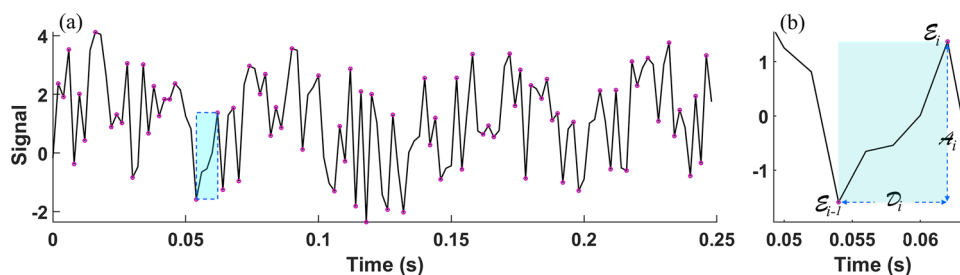


Fig. 1 Sample signal with extracted features for one segment included between consecutive local extrema, $\mathcal{E}_{i-1}$ and $\mathcal{E}_i$. Panel (a) shows an illustrative signal, with one highlighted segment. Panel (b) shows the

zoomed plot of the highlighted segment from panel (a), with marked extrema and the corresponding extracted features, i.e., the duration $\mathcal{D}_i$ and amplitude $\mathcal{A}_i$ of that segment

$([x(t_i), x(t_{i+1}), \dots, x(t_{i+m-1})])$. Two templates are considered a match if their distance is within the threshold r . To compute this, different measures such as Euclidean distance or Chebyshev distance can be used; the latter is more common:

$$d(\mathbf{T}_i^{(m)}, \mathbf{T}_j^{(m)}) = \max_{k=1, \dots, m} |T_i^{(m)}(k) - T_j^{(m)}(k)| \leq r. \quad (6)$$

B_i is the number of matches for all the template vectors of length m , i.e., the number of all indices $j \neq i$ where the difference between $\mathbf{T}_j^{(m)}$ and $\mathbf{T}_i^{(m)}$ is less than a given threshold, r , where $\mathbf{T}_i^{(m)}$ is the template of length m , starting at sample i :

$$B = \sum_{i=1}^{N-m} \sum_{j=i+1}^{N-m+1} \mathbb{I}(d(\mathbf{T}_i^{(m)}, \mathbf{T}_j^{(m)}) \leq r), \quad (7)$$

where $\mathbb{I}(\cdot)$ is the indicator function, which returns 1 if the condition is met and 0 otherwise. When counting matches for a given template, the template itself does not count (self-matches are excluded).

Similarly, A_i is the number of matches for the template vector of length $m + 1$, starting at the position i :

$$A = \sum_{i=1}^{N-m-1} \sum_{j=i+1}^{N-m} \mathbb{I}(d(\mathbf{T}_i^{(m+1)}, \mathbf{T}_j^{(m+1)}) \leq r). \quad (8)$$

The set of all possible templates is constructed by taking all the windows of length m (or $m + 1$) from the time series and computing the matching pairs.

The idea of sample entropy is based on finding the probability that a matching template of length m remains a match when extended to length $m + 1$. A lower SampEn value indicates a more predictable signal, while a higher value suggests increased complexity or randomness.

2.4 Duration, amplitude and their joint entropies

To quantify the complexity of the signal with the ExSEnt method, we compute the sample entropy of the sequence of extracted features: the durations of the intervals between each pair of consecutive extrema, $\{\mathcal{D}_k\}$, their net amplitude differences, $\{\mathcal{A}_k\}$, and the paired sequence of these features, $\{(\mathcal{D}_k, \mathcal{A}_k)\}$, as follows:

$$\mathcal{H}_{\mathcal{D}} = \text{SampEn}(m, r, \{\mathcal{D}_k\}), \quad (9)$$

$$\mathcal{H}_{\mathcal{A}} = \text{SampEn}(m, r, \{\mathcal{A}_k\}), \quad (10)$$

$$\mathcal{H}_{\mathcal{DA}} = \text{SampEn}(m, r, \{(\mathcal{D}_k, \mathcal{A}_k)\}). \quad (11)$$

These three entropy measures offer complementary perspectives on the complexity of the signal, capturing different aspects of its underlying dynamics. By analyzing these measures collectively, a more comprehensive characterization of the signal's dynamics is achieved, providing deeper insights into its intrinsic nature.

$\mathcal{H}_{\mathcal{D}}$ quantifies the degree of temporal variability of segments, capturing whether it follows an underlying pattern or exhibits irregular behavior. Similarly, $\mathcal{H}_{\mathcal{A}}$ measures the diversity of segments' amplitude variations, reflecting whether the profile of amplitude fluctuations is recurrent or more erratic.

Meanwhile, the joint entropy $\mathcal{H}_{\mathcal{DA}}$ characterizes the relation between these two features, assessing the extent to which these two sequences are independent. A higher $\mathcal{H}_{\mathcal{DA}}$ suggests a weaker coupling between duration and amplitude changes, indicating more independent or complex interactions between these features. Conversely, lower values suggests stronger synchronization, pointing to structured dependencies in the signal's temporal and amplitude modulations.

2.5 Computation of the joint entropy

To compute the joint entropy, the $\{\mathcal{D}\}$ and $\{\mathcal{A}\}$ sequences were first normalized to zero mean and unit standard deviation. Because these sequences are generated from different scales and units—time vs. amplitude—normalization is necessary to make them comparable before pairing. After that, we construct an $M \times 2$ matrix, where each row consists of a segment's duration and amplitude, and M is the number of extracted segments. At each step i , a sequence of m consecutive pairs is extracted and combined into a single vector, forming the i^{th} paired template:

$$\mathbf{P}_i^{(m)} = (\mathbf{p}_i, \mathbf{p}_{i+1}, \dots, \mathbf{p}_{i+m-1}), \quad \mathbf{p}_j = [\mathcal{D}_j, \mathcal{A}_j] \quad (12)$$

Where: $i, j \in \{1, 2, \dots, M - m - 1\}$. Similarly, template of length $m + 1$ is constructed as:

$$\mathbf{P}_i^{(m+1)} = (\mathbf{p}_i, \mathbf{p}_{i+1}, \dots, \mathbf{p}_{i+m}).$$

After obtaining the normalized paired templates, the remainder of the SampEn calculation follows the conventional procedure used in standard implementations. For each pair of normalized templates, $(\mathbf{P}_i^{(m)}, \mathbf{P}_j^{(m)})$, their similarity is evaluated based on Chebyshev distance (Eq. 6), the number of matched pairs is then computed as in Eqs. (7, 8). The sample entropy is thus calculated as in Eq. (5):

$$\mathcal{H}_{\mathcal{D}, \mathcal{A}} = -\log\left(\frac{A}{B}\right)$$

To compute the joint sample entropy of the paired set $(\mathcal{D}_i, \mathcal{A}_i)$, it is crucial to maintain consistency with the individual sample entropy evaluations over $\mathcal{D}_i$ and $\mathcal{A}_i$, i.e., $\mathcal{H}_{\mathcal{D}}$ and $\mathcal{H}_{\mathcal{A}}$. By preserving the same m —which translates to sequences with length $2 \times m$ in the paired scenario—the complexity of the joint system is assessed on the same scale as its individual components, making the comparison of results possible and meaningful.

The joint entropy provides insight into the mutual information between segment durations and amplitudes. If $\mathcal{H}_{\mathcal{D}, \mathcal{A}}$ is greater than or equal to the entropy of either durations or amplitudes, it suggests lower mutual information, implying that durations and amplitudes contain largely independent information. Conversely, if it is lower than both individual entropies, this suggests statistical dependence and coupling between the two variables, as some variability in one metric is already reflected in the other. This methodology allows for the quantification of the degree of temporal and amplitude-driven coupling between local extrema-based

segments within the time series, helping to better characterize the system's dynamical structure.

2.6 ExSEnt parameter choices

Throughout the following sections, all analyses were performed with the embedding dimension fixed at $m = 2$ (unless stated otherwise) and the tolerance parameter, r set to 0.2 of the time series standard deviation, consistent with widely adopted practice in the SampEn literature [17, 44, 68]. Longer embeddings require more data for meaningful estimation, hence our choice of m is also motivated by the need to ensure practical applicability in real time series. Nevertheless, the parameters m and r can be systematically explored and adjusted based on the available data length, the characteristics of the data, and the objective of the study.

On the other hand, in ExSEnt, SampEn is not applied to the original sample-by-sample time series, but to derived sequences formed by the extrema-based segment descriptors. Consequently, ExSEnt captures the structural information carried by the patterns of local extrema rather than the patterns of consecutive samples. Therefore, while we report SampEn on the time series using the same m as a baseline, the interpretation of m differs between the two computations. In this sense, the comparison to SampEn is intended to isolate the added value of replacing sample-by-sample templates with extrema-pattern templates, and to show how this substitution can disentangle temporal- versus amplitude-driven contributions that are conflated in the single SampEn value.

Moreover, ExSEnt can be applied in the same way to both discrete time maps and continuous time series, since it operates on the sampled trajectory through extrema segmentation and does not require any modification of the formulation across system types. In contrast, SampEn computation requires choosing a discrete sampling of the trajectory (e.g., an explicit downsampling factor), because its numerical value depends on it.

The only parameter specific to ExSEnt is the noise-threshold coefficient λ used in the segmentation step. To choose a reliable value, we performed simulations on synthetic signals with additive Gaussian noise and colored noise, which indicated that ExSEnt estimates are robust to the choice of λ provided that it remained below an application-dependent critical value (beyond which segmentation can fail or become overly coarse). In our experiments, stable behavior was typically observed for λ in the range [0.01, 0.05] (Supplementary Information, S1). Based on these results, we set $\lambda = 0.01$ throughout our analyses as a conservative value that improves noise tolerance without suppressing meaningful extrema.

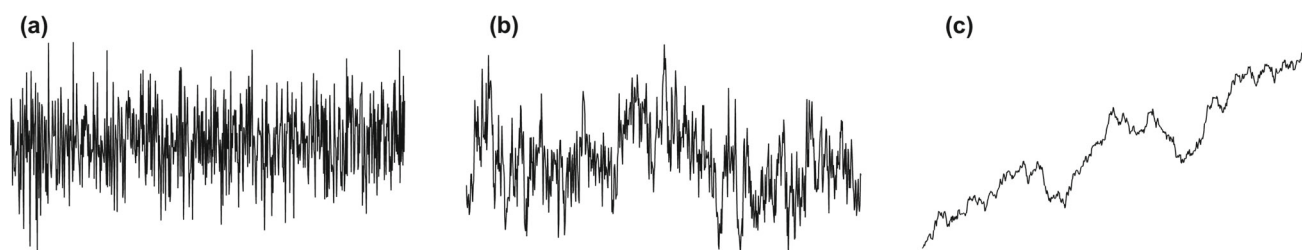


Fig. 2 Representative stochastic signals. Panels (a) to (c) show Gaussian noise, Pink noise, and Brownian motion, respectively. The signals were generated according to the description presented in section A.1 and are plotted here over $N = 700$ samples

Table 1 ExSEnt vs. SampEn estimates of stochastic signals.

Signal	$\mathcal{H}_{\mathcal{D}}$	$\mathcal{H}_{\mathcal{A}}$	$\mathcal{H}_{\mathcal{DA}}$	SampEn
Gaussian	0.736 ± 0.010 CV = 0.014	1.348 ± 0.006 CV = 0.004	0.971 ± 0.009 CV = 0.009	2.185 ± 0.005 CV = 0.002
Pink	0.934 ± 0.010 CV = 0.011	1.479 ± 0.006 CV = 0.004	1.062 ± 0.010 CV = 0.009	1.647 ± 0.037 CV = 0.023
Brownian	1.100 ± 0.014 CV = 0.013	1.449 ± 0.009 CV = 0.006	1.067 ± 0.013 CV = 0.012	0.048 ± 0.021 CV = 0.468

All computations were performed over $N = 20000$, with $m = 2$. We report the mean $\pm$ SD along with the Coefficient of Variation (CV) over 100 random realizations

3 Results

In this section, we present the results obtained by applying the ExSEnt method to synthetic and real benchmark data from various origins: stochastic signals, widely studied chaotic systems—the Logistic Map (LM), the Rössler System (RS), and the Rulkov map (RM)—and two human movement datasets—electromyography and inertial time series. In all cases, results are compared with corresponding SampEn estimates. We also study the effect of data length on ExSEnt measures.

All entropy values reported here are interpreted as estimator-specific complexity indices under these fixed pre-processing and parameter settings, and are used primarily for within-study comparisons (across conditions or parameter sweeps) rather than as absolute, universally comparable scales.

3.1 ExSEnt for representative stochastic signals

We computed the ExSEnt metrics for Gaussian noise, Pink noise, and Brownian motion (Fig. 2). Table 1 summarizes the statistics of the ExSEnt measures computed over 100 runs, along with the SampEn.

The results revealed distinct patterns across the signals. For Gaussian noise, the amplitude entropy $\mathcal{H}_{\mathcal{A}}$ was higher than the other ExSEnt metrics, reflecting substantial randomness in the signal's amplitude fluctuations, while the comparably lower $\mathcal{H}_{\mathcal{D}}$ suggested less variability in extrema-based segment durations. Pink noise exhibited both higher duration and amplitude entropies compared to Gaussian noise, and its resulting joint entropy was slightly higher than that of Gaussian noise. In the case of Brownian motion,

the analysis showed the highest duration entropy compared to the other signals, demonstrating the large variability in the lengths of the segments, while its amplitude entropy remained marginally lower than Pink noise.

Since spectral coloring mainly modifies temporal correlation structure, trends of $\mathcal{H}_{\mathcal{A}}$ across Gaussian, pink, and Brownian signals need not be monotonic. We observed that $\mathcal{H}_{\mathcal{A}}$ was highest for Pink noise and lowest for Gaussian noise; nonetheless, they were all within a similar range, $1.348 - 1.479$, with very small variation across simulation runs. whereas $\mathcal{H}_{\mathcal{D}}$ was expected to be more sensitive to changes in correlation via the extrema spacing, and this was also reflected in the $\mathcal{H}_{\mathcal{D}}$ values, where there was a more significant difference between them; Gaussian had the lowest value and the Brownian manifested the highest level of temporal entropy.

Comparing these estimates with the SampEn results demonstrated the extended ability of ExSEnt metrics to quantify the complexity of the signal. With respect to $\mathcal{H}_{\mathcal{DA}}$, for both Gaussian and Pink noise, SampEn produces higher entropies than joint ExSEnt, while the SampEn for the Brownian is considerably small—much smaller than its $\mathcal{H}_{\mathcal{DA}}$ —with a relatively high CV. Hence, SampEn did not reflect the complex dynamics of Brownian motion, while ExSEnt captured it in all three measurements.

These findings confirmed that the ExSEnt method was sensitive to the underlying dynamics of different stochastic signals. The ability to capture both the individual and joint complexities of segment durations and amplitudes provided valuable insights into the temporal and amplitude-driven variability of these signals.

3.2 ExSEnt for representative nonlinear dynamical systems

We analyzed three low-dimensional nonlinear dynamical systems, the Logistic Map (LM), the Rössler System (RS), and the Rulkov Map (RM), using ExSEnt measures. By quantifying the distribution and sequencing of extrema, ExSEnt discriminates periodic, near-periodic, and chaotic regimes, and highlights transitions such as period-doubling cascades and intermittency. These metrics thus provide complementary sensitivity to changes across parameter space and yield interpretable signatures that deepen our understanding of each system's dynamics. The control parameters r , c , and σ for LM, RS, and RM, respectively, govern the systems' transition from periodic behavior to chaos, making them ideal test cases for evaluating ExSEnt measures.

Logistic map The LM is a one-dimensional discrete-time quadratic map (A.2.1) that serves as a model of nonlinear behavior and chaos via the period-doubling route [65].

$$x_{n+1} = rx_n(1 - x_n), \quad (13)$$

We computed the ExSEnt metrics along with the SampEn, over $r \in [2.9, 4]$. Panel (a) in Fig. 3 shows the bifurcation diagram, color-coded based on normalized $\mathcal{H}_{\mathcal{DA}}$, highlighting regions of varying complexity. Panel (b) shows the ExSEnt measures $\mathcal{H}_{\mathcal{D}}$, $\mathcal{H}_{\mathcal{A}}$, and $\mathcal{H}_{\mathcal{DA}}$, and panel (c) shows the SampEn metric. While $\mathcal{H}_{\mathcal{DA}}$ had a similar pattern to SampEn, we observed that $\mathcal{H}_{\mathcal{D}}$ and $\mathcal{H}_{\mathcal{A}}$ had distinct values, with $\mathcal{H}_{\mathcal{A}}$ being larger than $\mathcal{H}_{\mathcal{D}}$ for most of the range of r , except the intermittent windows. By intermittent, we refer to regimes with long, nearly regular epochs interrupted by irregular bursts near the transition to chaos [69]. In particular, as r is increased from the periodic regime, $\mathcal{H}_{\mathcal{D}}$ exceeded zero only for $r \geq 3.676$, while $\mathcal{H}_{\mathcal{DA}}$ and $\mathcal{H}_{\mathcal{A}}$ started to exhibit non-zero values earlier, for $r \geq 3.544$. This indicates a distinction between the behaviors of duration and amplitude in the interval [3.544 3.676]: the system exhibits complex dynamics arising from variability in the patterns of amplitude, while the duration entropy remains zero because the signal first-order difference alternates in sign at every sample, maintaining a constant pattern of durations between successive extrema (Fig. 3d). Panels (e) and (f) show the LM time series and its phase space, with lag $\tau = 1$, for two other sample control parameter values. For $r = 3.742$, while the amplitude and joint entropies were close to zero, the duration entropy was 0.406. This indicates that the magnitudes of successive extrema excursions are essentially repetitive (hence $\mathcal{H}_{\mathcal{A}} \approx 0$ and $\mathcal{H}_{\mathcal{DA}} \approx 0$), whereas the extrema-to-extrema durations still exhibit variability when assessed with short templates ($m = 2$). Notably, increasing the embedding dimension to $m = 3$ drives $\mathcal{H}_{\mathcal{D}}$ to zero as

well, suggesting that the apparent timing variability has a short, finite-memory structure: once a longer temporal context (three consecutive extrema-defined segments) is taken into account, the temporal sequence becomes deterministic. This dissociation implies that the temporal component requires a larger effective embedding dimension (longer memory) than the amplitude component in this regime, i.e., temporal variability is structured and resolves when conditioned on a longer extrema-pattern context. Panel (f) shows a regime where the entropies of amplitude and durations were both higher than their joint entropy.

To assess the extent to which ExSEnt measures are sensitive to the deterministic structure of the time series in illustrative cases, we compared the ExSEnt metrics obtained for $r = 3.600$ and $r = 3.970$ with corresponding distributions of 1000 IAAFT surrogates [70], which preserved the amplitude distribution and spectral content of the original signal but destroyed higher-order temporal dependencies. For $r = 3.600$, where the signal derivative alternates in sign at each sample, the surrogate statistics (mean $\pm$ SD) were $\mathcal{H}_{\mathcal{D}} = 0.000 \pm 0.000$, $\mathcal{H}_{\mathcal{A}} = 0.397 \pm 0.002$, and $\mathcal{H}_{\mathcal{DA}} = 0.179 \pm 0.001$. Thus, the original value showed no significant departure from surrogates for $\mathcal{H}_{\mathcal{D}}$, whereas the original values of $\mathcal{H}_{\mathcal{A}}$ and $\mathcal{H}_{\mathcal{DA}}$ were lower than the corresponding surrogate means by approximately 85 and 89 SD, respectively. For $r = 3.970$, the surrogate statistics were $\mathcal{H}_{\mathcal{D}} = 0.731 \pm 0.009$, $\mathcal{H}_{\mathcal{A}} = 0.691 \pm 0.010$, and $\mathcal{H}_{\mathcal{DA}} = 0.764 \pm 0.007$, whereas the original values were lower than the corresponding surrogate means by approximately 28, 16, and 65 SD, respectively. Importantly, all ExSEnt metrics, except $\mathcal{H}_{\mathcal{D}} \approx 0$ for $r = 3.600$, were lower than every corresponding surrogate realization, indicating a clear departure from the surrogate distributions.

Rössler system The RS is a simple three-dimensional continuous-time dynamical model that exhibits chaos (A.2.2) and is widely used as a minimalist benchmark in nonlinear dynamics [66].

$$\begin{aligned} \frac{dx}{dt} &= -y - z, \\ \frac{dy}{dt} &= x + ay, \\ \frac{dz}{dt} &= b + z(x - c). \end{aligned} \quad (14)$$

The bifurcation diagram of the RS, color-coded by the normalized $\mathcal{H}_{\mathcal{DA}}$, effectively highlights chaotic intervals versus intermittent and periodic regimes (Fig. 4, panel a). Among the ExSEnt metrics, the duration-based entropy was higher than the other two (panel b). For intermittent and periodic parameter values, $\mathcal{H}_{\mathcal{A}}$ and $\mathcal{H}_{\mathcal{DA}}$ were relatively low (< 0.1). In

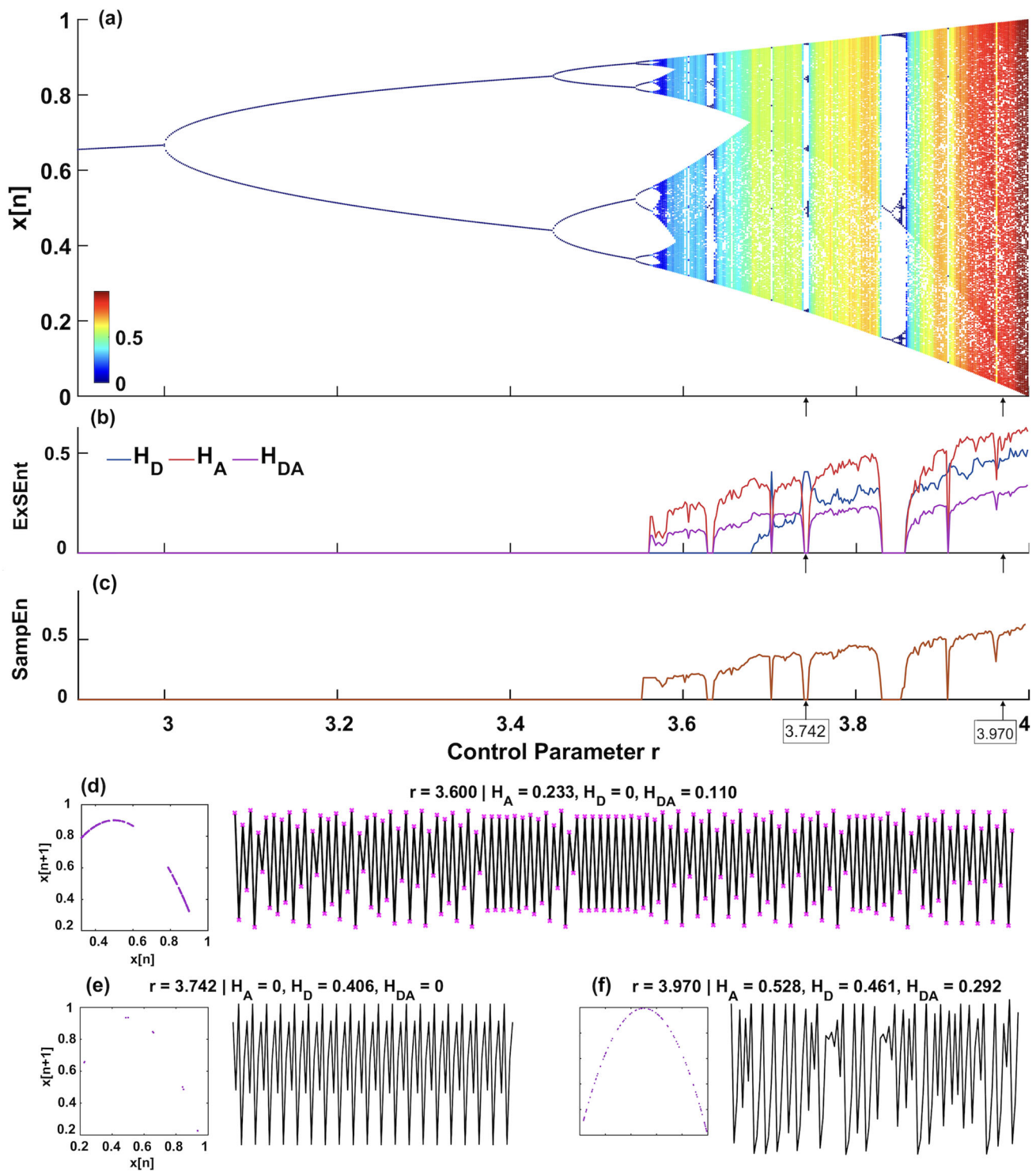


Fig. 3 Evaluation of the Logistic map with ExSEnt. **a** Bifurcation diagram of the logistic map, color-coded by normalized joint entropy $\mathcal{H}_{DA}$. **b** ExSEnt values for segment duration ($\mathcal{H}_D$), net amplitude ($\mathcal{H}_A$), and joint entropy ($\mathcal{H}_{DA}$); and **c** sample entropy (SampEn), all as a function of the control parameter r . Panel **(d)** shows the system's dynamics at lag-1 for $r = 3.600$, revealing a rich range of states also evident in the corresponding time series (right). The signal first-order

difference alternates in sign at every sample; hence, every sample is a local extremum (magenta crosses). This illustrates how examining complexity from both temporal and amplitude perspectives enhances our understanding of the system's behavior. Panels **(e)** and **(f)** depict the phase-space and time series of the map for the specified r values along with the corresponding ExSEnt measures. All metrics were computed over $N = 10^4$ data points

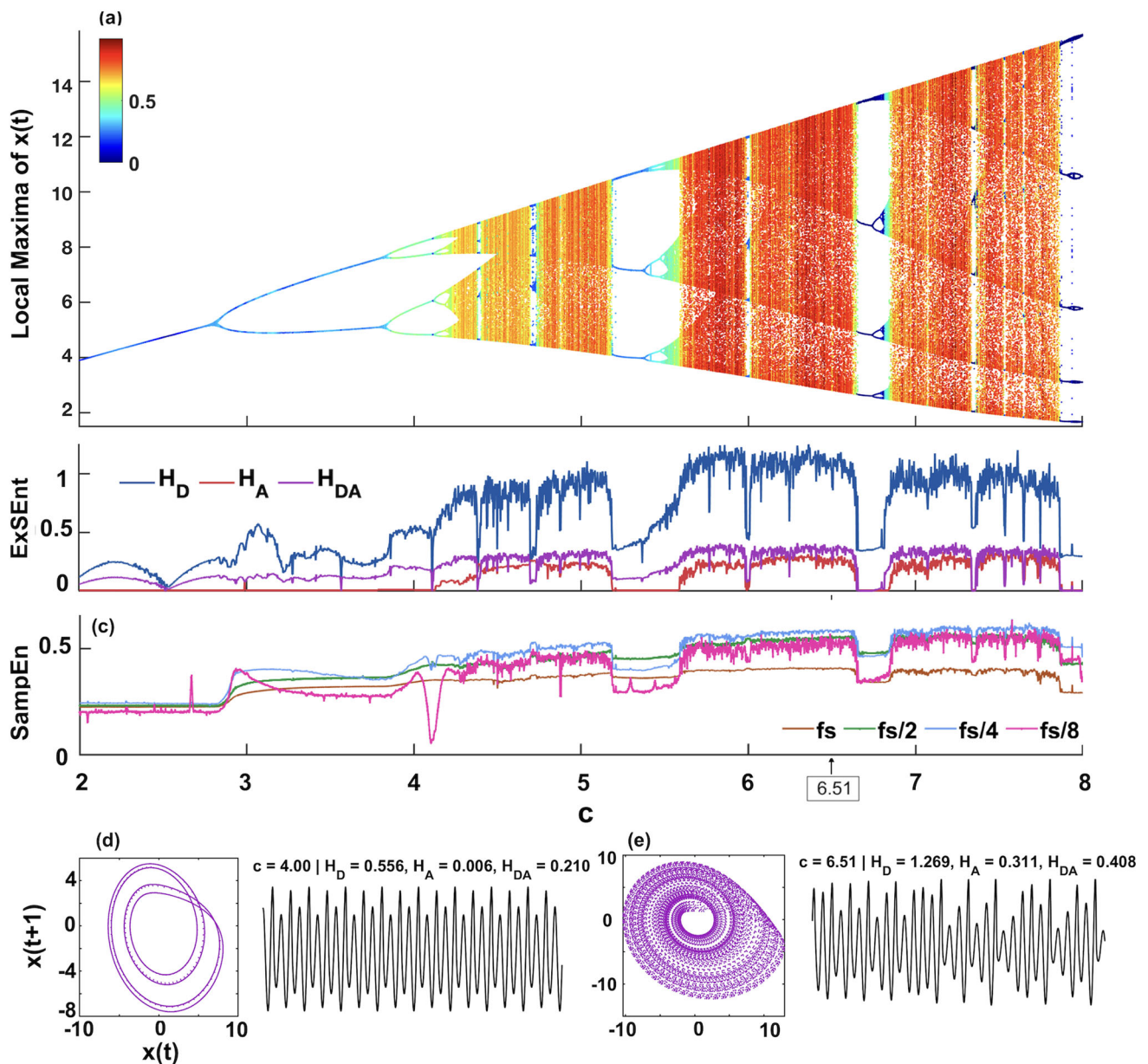


Fig. 4 Bifurcation diagram and entropy measures of the Rössler system. Panel (a) shows the bifurcation diagram color-coded by the normalized joint entropy $\mathcal{H}_{DA}$. The ExSEnt metrics $\mathcal{H}_D$, $\mathcal{H}_A$, and $\mathcal{H}_{DA}$ are depicted as a function of the control parameter c in panel (b). The parameters used were $a = 0.2$ and $b = 0.2$, with sampling frequency $f_s = 10$, $N = 10^4$ data points, and $c \in [2, 8]$. Sample entropy of the signal over the given range of the control parameter is also shown for the raw signal,

along with its three down-sampled versions (panel c). Panels (d) and (e) show the phase space and time series for $c = 4.00$ and $c = 6.51$, at two different regimes—periodic and chaotic, respectively—with their ExSEnt metrics. The ExSEnt metric provides a measure of information-theoretic complexity for each case. The selected values of c highlight transitions between different dynamical regimes

particular, $\mathcal{H}_A$ is almost zero for wide ranges of the control parameter, whereas $\mathcal{H}_D$ remained consistently high (≥ 0.5) across most of the examined range.

Because the RS is a continuous-time system, applying SampEn directly to the raw signal does not provide a fully fair comparison with ExSEnt. For this reason, in addition to SampEn computed from the raw signal, we also evaluated

SampEn after downsampling the signal to $\frac{1}{2}$, $\frac{1}{4}$, and $\frac{1}{8}$ of the original sampling frequency. The results showed that the $fs/8$ case produced the pattern most similar to the ExSEnt measures, with the closest resemblance to $\mathcal{H}_D$. Panels (d) and (e) illustrate two distinct regimes: periodic and chaotic. For $c = 4$, $\mathcal{H}_A = 0.006$ whereas $\mathcal{H}_D = 0.556$ and $\mathcal{H}_{DA} = 0.210$. Setting $m = 3$ lowered ExSEnt estimates to $\mathcal{H}_D =$

0.172 and $\mathcal{H}_{\mathcal{DA}} = 0.082$. Panel (e) shows the phase-space and the time series of the RS for $c = 6.51$, which corresponds to the chaotic regime.

For the case $c = 4$, $\mathcal{H}_{\mathcal{A}}$ close to zero, indicated that extrema-to-extrema net magnitude differences were highly stereotyped and that an embedding of $m = 2$ was already enough to capture this near-deterministic amplitude structure. In contrast, the duration sequence exhibited a longer-range repeating template, i.e., a period-3 structure in the extrema spacing. Hence, $m = 2$ templates do not fully represent the underlying repeating pattern, whereas increasing to $m = 3$ aligns the template length with the dominant motif and reduces the estimated temporal and joint entropies. This example illustrates that temporal versus amplitude variability can have different effective memory lengths, and that the ExSEnt decomposition can reveal which component carries the structured dependence. Patterns like this do not mean that the system has two distinct physical dimensions. Rather, it implies that $\mathcal{D}$ and $\mathcal{A}$ capture different aspects of the system's dynamics and variability, and they could require templates with different lengths to capture the predictability of the system.

Rulkov map RM is a discrete-time system, modeling neuronal bursting dynamics, with two variables x_n and y_n generating fast and slow dynamics:

$$\begin{aligned}x_{n+1} &= \frac{\alpha}{1 + x_n^2} + y_n, \\y_{n+1} &= y_n - \mu(x_n - \sigma).\end{aligned}\quad (15)$$

Here, α controls excitability, the small parameter μ enforces the separation between fast and slow timescales, and σ acts as the bifurcation parameter driving transitions between dynamical regimes (see A.2.3 for the full description).

RM generates key firing patterns, such as silence, tonic spiking, bursting, mixed-mode oscillations, and chaos, via slow-fast dynamics [67]. Being computationally lightweight and physiologically interpretable makes it widely used for large-scale network simulations and bifurcation-based studies of excitability and bursting regimes. The bifurcation diagram of the RM for the control parameter σ , color-coded by the joint ExSEnt, highlights chaotic bursting intervals versus intermittent and periodic regimes (Fig. 5a). Comparing panels (b) and (c), although the overall pattern of SampEn almost resembled the smoothed $\mathcal{H}_{\mathcal{D}}$ and $\mathcal{H}_{\mathcal{DA}}$ (panel c), the ExSEnt measures exhibited richer variability (panel b), providing a more nuanced characterization of the signal's complexity. While $\mathcal{H}_{\mathcal{A}}$ was relatively smooth for $\sigma < -0.3650$, $\mathcal{H}_{\mathcal{D}}$ (and, to a smaller degree, $\mathcal{H}_{\mathcal{DA}}$) showed step-like rises at $\sigma \in \{-1.0675, -0.8650, -0.6025\}$, whereas SampEn remained smooth. Specifically, while $\mathcal{H}_{\mathcal{D}}$ showed

a smooth decreasing trend as σ decreased, the other two ExSEnt metrics showed a step-like increasing trend at this interval, depicting a dissociation between the complexity of $\mathcal{D}$ and $\mathcal{A}$. By contrast, SampEn failed to capture this complexity, collapsing to near-zero values as σ approached -1.5 .

Panels (d) and (e) focus on two parameter values within this step-like region. Although the time series and phase portraits for $\sigma = -0.605$ and $\sigma = -0.603$ (this value was -0.6025 which is rounded to three decimals in panel e) appeared similar, and the entropies of $\mathcal{A}$ and $\mathcal{DA}$ were comparable, we observed a 42.8% increase in the entropy of $\mathcal{D}$ in the latter case compared to the former.

Another noteworthy interval was $\sigma \in [0.773, 0.798]$, where $\mathcal{H}_{\mathcal{A}}$, $\mathcal{H}_{\mathcal{DA}}$, and SampEn were near zero (≤ 0.07), while $\mathcal{H}_{\mathcal{D}} \geq 0.4$. We also inspected $\sigma = 0.790$ (periodic; panel f) and $\sigma = 0.8500$ (chaotic), whose phase spaces corroborated these classifications. For $\sigma = 0.790$ and $m = 2$, $\mathcal{H}_{\mathcal{D}}$ was considerably higher than the other ExSEnt metrics, raising the question of whether the effective embedding dimension of $\mathcal{D}$ was higher than 2. The term effective embedding dimension refers to the minimum embedding dimension required to unfold the underlying dynamics without introducing false neighbors, i.e., the dimension at which the reconstructed attractor preserves the system's topology [71]. Increasing the embedding dimension to $m = 3$ yielded $\mathcal{H}_{\mathcal{D}} = 0$, highlighting another example where the effective embedding dimension is higher for $\mathcal{D}$ than for $\mathcal{A}$.

Figs. 6a and b provide a closer look at the $\sigma = -0.6050$ and -0.6025 phase spaces. Over long segments, both parameters showed similar patterns at the three-dimensional lag one phase-space (panels a and b, left plots). However, restricting the visualization to a short segment and explicitly connecting consecutive states reveals differences in the transition dynamics: for $\sigma = -0.6025$, the trajectory of the system yielded a denser web of short-range paths and a higher degree of state-to-state variability than for $\sigma = -0.6050$. In contrast, $\sigma = -0.6050$ showed more repeated, coherent sweeps with fewer variability in the transitions from state to state, consistent with a more regular temporal organization.

This qualitative change aligns with the increase in the $\mathcal{H}_{\mathcal{D}}$, which is sensitive to variability in the sequence of inter-extrema durations. Importantly, this effect can be partially decoupled from changes in the amplitude organization captured by $\mathcal{H}_{\mathcal{A}}$: two trajectories may occupy a similar region of state space, i.e., the same amplitude sets, while differing in the order of traversing that space. Therefore, decomposing ExSEnt into duration- and amplitude-driven contributions provides a more mechanistic interpretation, separating temporal from amplitude-driven irregularities of successive excursions, and clarifying why subtle parameter changes can increase dynamical complexity even when the global attractor shape appears comparable.

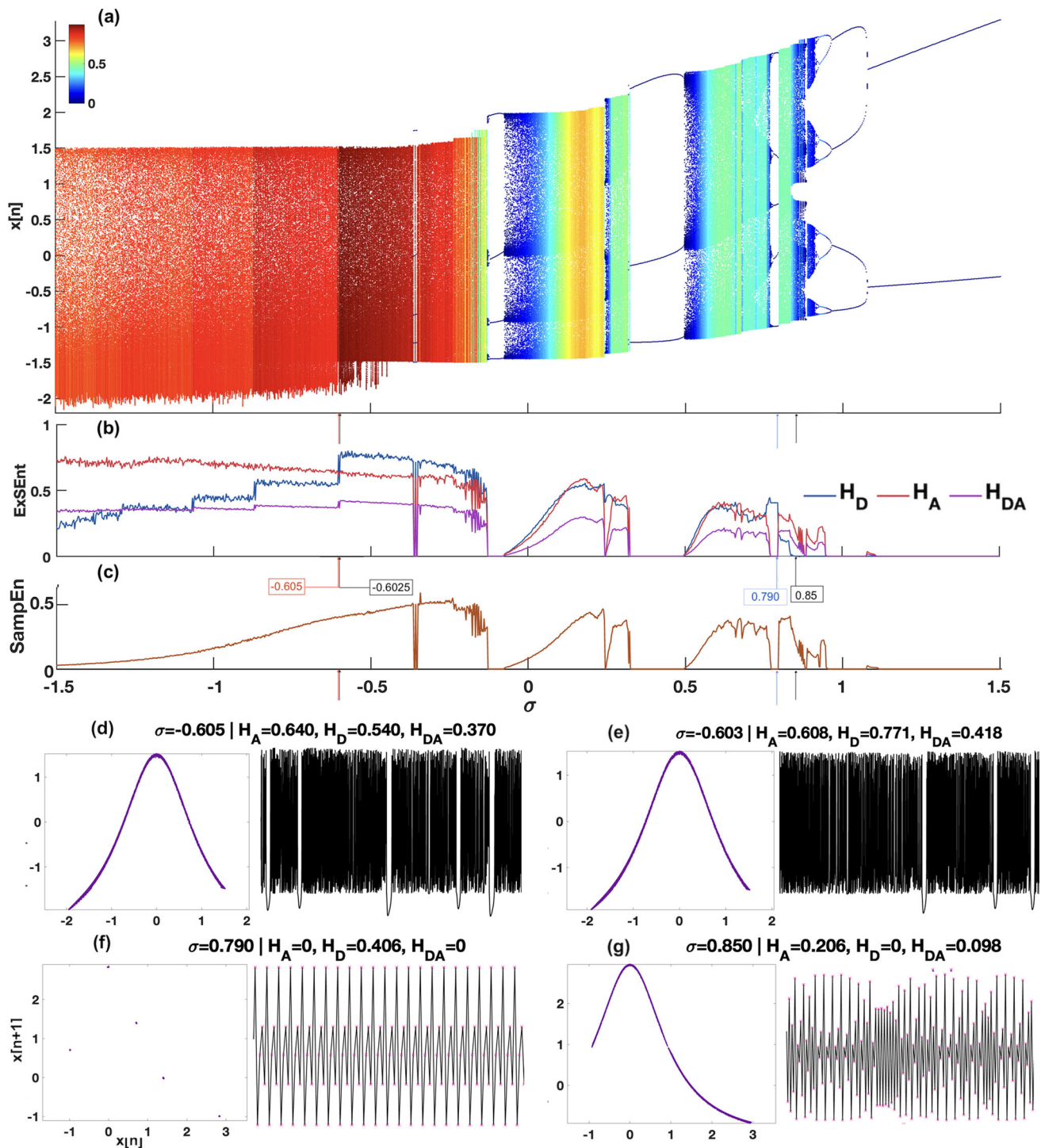


Fig. 5 Bifurcation diagram and entropy measures of the Rulkov map. Panel (a) shows the bifurcation diagram color-coded by the normalized joint entropy $\mathcal{H}_{DA}$. The ExSEnt metrics as a function of the control parameter σ are depicted in panel (b). The parameters used were $\alpha = 4.3$, $\mu = 0.001$, $N = 1.5 \times 10^4$ data points, and $\sigma \in [-1.5, 1.5]$, with initial values $x_0 = -1.5$, $y_0 = -2$. Sam-

ple entropy over the given range of the control parameter is also shown (panel c). Panels (d)–(g) show the phase space and time series for $\sigma \in \{-0.6050, -0.6025, 0.7900, 0.8500\}$, illustrating different bursting regimes (periodic, intermittent and chaotic), with their corresponding ExSEnt metrics

Panel (c) in Fig. 6 displays three representations of the phase space for the RM system at $\sigma = 0.790$. The left plot

illustrates the phase space, after discarding the initial 500 points and over $N = 15000$ data points. This plot hence

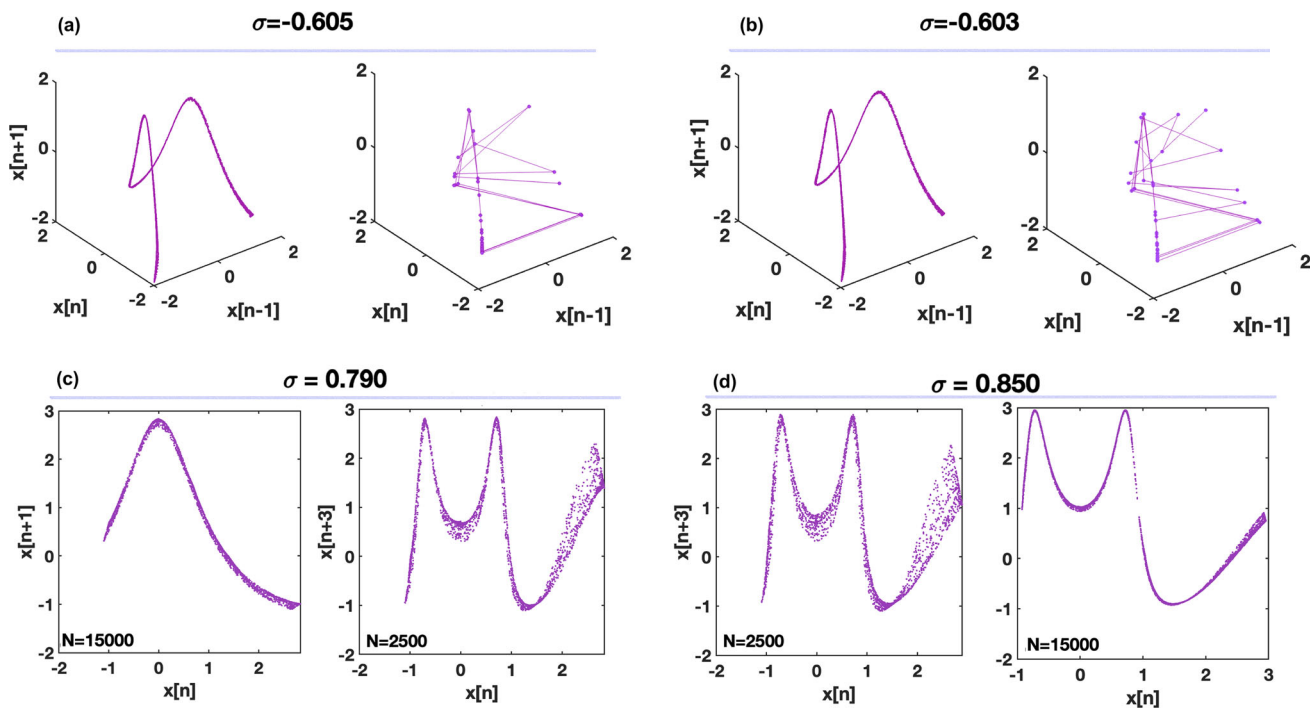


Fig. 6 Phase space for selected σ values of the Rulkov map. All the parameters are the same as Fig. 5. Panels (a) and (b) show the phase space of RM for $\sigma \in \{-0.6050, -0.6025\}$, with lag $\tau = 1$. The left plots show the phase space over $N = 15000$ points, while the right plots show the phase space over $N = 100$ points with lines connecting consecutive states. Panel (c) shows the phase space of RM for $\sigma = 0.790$.

The first plot on the left shows the phase space over $N = 15000$ data points. The right panel for $\sigma = 0.790$ shows the phase space with $\tau = 3$ for $N = 2500$. Panel (d) shows the phase space of RM for $\sigma = 0.850$ with $\tau = 3$, once over $N = 2500$ and then over $N = 15000$. The first three plots in this row are produced by discarding only the first 500 points

included the transient phase, which exhibits similarity to the $\tau = 1$ phase space observed at $\sigma = 0.850$ (Fig. 5, panel g). However, when discarding the transient phase, these two parameter values yielded strikingly different dynamics as assessed by our ExSEnt metrics: periodic behavior with only $\mathcal{H}_D$ being non-zero for $\sigma = 0.790$, versus chaotic behavior with $\mathcal{H}_D = 0$ and non-zero $\mathcal{H}_A$ and $\mathcal{H}_{DA}$. To better understand the underlying dynamics, we examined the phase space for the transition phase. Specifically, we discarded the first 500 data points and then computed the ExSEnt measures over the first 2500 data points. For $\sigma = 0.790$, the ExSEnt measures of the transient phase were $\mathcal{H}_A = 0.423$, $\mathcal{H}_D = 0.311$, $\mathcal{H}_{DA} = 0.219$, which were in a similar range to the ExSEnt metrics for $\sigma = 0.850$ where we had $\mathcal{H}_A = 0.382$, $\mathcal{H}_D = 0.343$, and $\mathcal{H}_{DA} = 0.200$. This demonstrated that the system manifests a similar complexity pattern in the transition phase for these two different control parameter values. The middle sub-panels in the second row show the transition phase with $\tau = 3$ as described. The right plot on panel (d) shows the phase space at $\tau = 3$ for 15000 data points after the transition period is discarded.

These findings highlight the capability of ExSEnt to distinguish different dynamical regimes and capture subtle complexity variations that are not fully resolved by conven-

tional entropy measures. By effectively characterizing both individual and joint entropy contributions, ExSEnt provides a nuanced understanding of the interplay between temporal and amplitude fluctuations across different system states. Various methods and algorithms are proposed to determine the optimal embedding delay and dimension, such as the use of mutual information for estimating time delays [72] and different methods for embedding dimension calculation such as the false nearest neighbors approach [71], or Cao's method, which evaluates the independence of reconstructed vectors in the phase space to identify the minimum sufficient dimension [73]. Our observations suggest that when temporal and amplitude-driven patterns of a signal are analyzed separately, they may exhibit distinct characteristics across different lags and embedding dimensions. In particular, the dimensionality of the temporal and amplitude components of a signal may not necessarily coincide. While a detailed exploration of these phenomena lies beyond the scope of the present study, feature-decomposed approaches to signal complexity could offer new insights into the dimensionality and phase space characterization of complex systems.

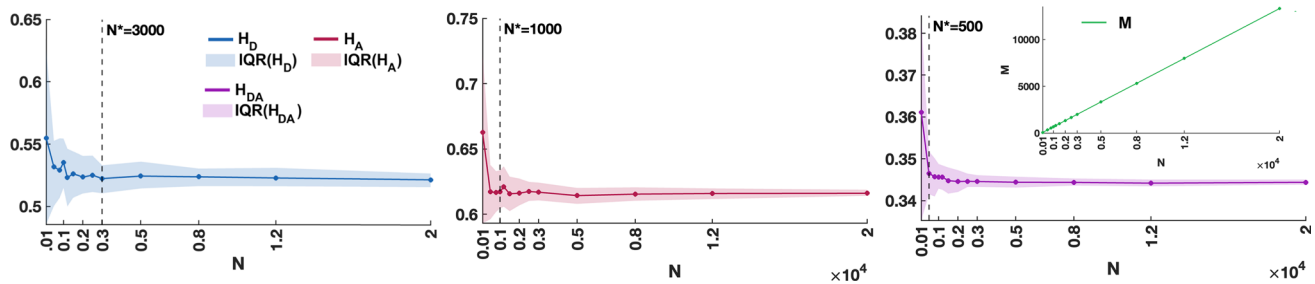


Fig. 7 ExSEnt convergence properties over the number of samples N on the logistic map with $r = 4$. For each N , the metrics were computed over 100 realizations with random initial conditions. The mean of each ExSEnt metric is plotted with a shaded IQR band, as indicated

3.3 Convergence properties of ExSEnt measures

The convergence of a metric with respect to data length is critically important, as it ensures consistent characterization of dynamical systems across varying observational scales and prevents misinterpretation of emergent features that may be finite-size effects. We investigated the convergence of ExSEnt metrics as a function of N using the logistic map. Because the number of extracted segments M determines the lengths of the $\mathcal{D}$ and $\mathcal{A}$ sequences, we also quantified how M scales with N . The logistic map provides an ideal testbed for convergence analysis due to its measure-preserving properties in the chaotic regime, at $r = 4$ [74], where invariant statistics persist across distinct trajectories generated by varying initial conditions. This invariance reflects the ergodic nature of the map, whereby time averages converge to ensemble averages under the invariant measure [75, 76]. We fixed $r = 4$ and computed the ExSEnt metrics across 100 realizations per N using random initial conditions, with the mean stability curve and its IQR depicted as a shaded region in Fig. 7.

As N increased, the entropy measures converged to stable values, indicating that the method yielded consistent estimates for sufficiently long time series. A dashed black vertical line indicated the smallest N at which the metric entered an acceptable stability range, defined as an IQR band whose half-width did not exceed 2.5% of the metric's final value. Under this criterion, stability was reached at $N = 3000$ for $\mathcal{H}_{\mathcal{D}}$, at $N = 1000$ for $\mathcal{H}_{\mathcal{A}}$ and at $N = 500$ for $\mathcal{H}_{\mathcal{DA}}$. The inset shows the mean number of extracted segments M as a function of the number of samples N , indicating that M increases linearly with N . Although $N \approx 3000$ was a reasonable series length for the LM, convergence depends on the relationship between the sample size N and the number of extracted segments M , which varies with the signal's characteristics. For the LM, the M vs. N curve yielded $M \approx 0.67 \times N$. As expected, M depended on the signal's frequency content; thus, the minimum N ensuring stable esti-

mates is data-dependent and must be determined for each dataset. Ultimately, M is the key determinant of data sufficiency.

3.4 ExSEnt for biosignals

Nonlinear metrics play a critical role in evaluating biosignals because they quantify dynamical properties—such as complexity and pattern variability—that linear statistics cannot capture [7, 25, 77]. In particular, entropy-based measures summarize uncertainty and temporal organization. They can serve as candidate biomarkers when standardized across cohorts and conditions, being sensitive to pathology-related alterations while robust to moderate noise and amplitude scaling [25, 78]. We extracted ExSEnt measures from two real datasets: surface EMG recordings and inertial measurements.

EMG signal analysis We applied ExSEnt to EMG signals recorded from the forearms of 19 healthy subjects during a finger-pinching task. We evaluated the metrics in three movement phases: (1) a 2-second baseline interval, during which the subjects rested their arms on the table while awaiting the go-cue command; (2) a 2-second window during the execution of the movement; and (3) the entire 7-second trial (Fig. 8). We applied the ExSEnt measures to band-pass filtered signals in the 100–200 Hz frequency range, as within this frequency band the 2-second analysis window provides a sufficiently large number of extrema (M) to ensure reliable estimation. We observed that the movement decreased $\mathcal{H}_{\mathcal{A}}$ considerably (−23.95%), while $\mathcal{H}_{\mathcal{D}}$ showed only a slight reduction (−0.92%), indicating regularized amplitude modulation with substantially smaller decline in temporal complexity. The data and the preprocessing steps are described in [79, 80].

The EMG signals during the motor task revealed that movement intervals exhibit lower amplitude complexity but relatively stable temporal entropy compared to the baseline.

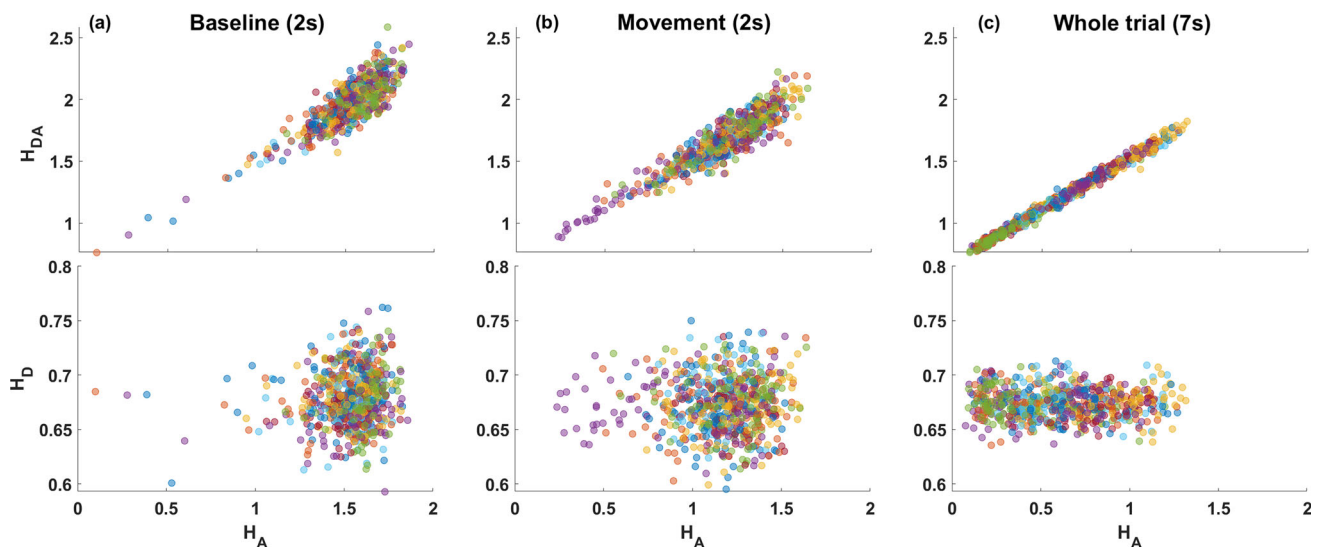


Fig. 8 ExSEnt metrics over surface EMG during a finger-pinching task: movement reduced amplitude complexity but preserved temporal entropy. Panel (a), top, shows joint ExSEnt ($\mathcal{H}_{DA}$) vs. amplitude entropy ($\mathcal{H}_A$) during a 2 s baseline interval; the bottom scatter plot shows duration entropy ($\mathcal{H}_D$) vs. $\mathcal{H}_A$ for the same interval. Panel (b),

top and bottom, is analogous to panel (a) but for a 2 s window extracted from the movement period. Panel (c), top and bottom, shows the same metrics over the entire 7 s task window. All metrics were computed over the 100–200 Hz band. Single-trials are color-coded consistently per subject

A strong linear correlation was observed between the $\mathcal{H}_A$ and $\mathcal{H}_{DA}$, suggesting that amplitude variations are the primary contributors to the joint amplitude-temporal EMG complexity. At the baseline, the electrodes on the arm capture the environmental noise and the tonic muscle activity at rest. The relatively higher entropy of manifested amplitudes during the resting state arises from these random signals, with small but irregular magnitudes. With movement initiation, the amplitude of neural activity increases considerably; synchronized neural populations fire with reduced variability performing the intended movement, which leads to the reduction in $\mathcal{H}_A$. This suggests that in monitoring the movement disorders, a candidate biomarker, which can potentially increase the precision of pathology detection and therapy improvement measurement, is to concentrate on the magnitude of the neural activities, as this could yield a sharp, mechanistically interpretable metric for prognosis.

Ankle motion acceleration in Parkinson's disease The Daphnet Freezing of Gait dataset documents a wearable assistant for Parkinson's disease that uses on-body accelerometers to monitor movement, detect freezing of gait (FOG), and deliver rhythmic auditory cues to help patients resume walking [81]. Ten PD patients performed laboratory walking tasks; eight patients exhibited FOG, and FOG events were verified from synchronized video recordings [81].

We analyzed vertical ankle acceleration (VAA) using the standard locomotor (0.5–3 Hz) and freeze (3–8 Hz) bands; the Freeze Index (FI)—the 3–8 Hz/0.5–3 Hz power ratio—has

Table 2 Statistical comparison (p -values) between NFG and FOG across frequency bands (two-sided Wilcoxon rank-sum test) for ExSEnt metrics over VAA from the Daphnet dataset (ns: nonsignificant)

Metric	0.5–3 Hz	3–8 Hz
$\mathcal{H}_D$	ns	0.038
$\mathcal{H}_A$	0.002	<0.001
$\mathcal{H}_{DA}$	0.015	<0.001
SampEn	<0.001	0.001

been validated with ankle/shank sensors in lab and ambulatory settings [82, 83]. We computed the three ExSEnt metrics ($\mathcal{H}_A$, $\mathcal{H}_D$, $\mathcal{H}_{DA}$) in both bands for no-freezing gait (NFG) and FOG windows. NFG windows contained no freezing episodes, whereas FOG windows mainly ($\geq 70\%$) consisted of the freezing state. All windows were at least 30s long. ExSEnt differed significantly between NFG and FOG for $\mathcal{H}_A$ and $\mathcal{H}_{DA}$ in the 0.5–3 Hz band and for all three metrics in the 3–8 Hz band (Table 2). For comparison, SampEn on the same signals also discriminated NFG from FOG (Fig. 10). However, ExSEnt resolved band-specific structure: in the 0.5–3 Hz band, duration entropy did not differ significantly between NFG and FOG, and the other two metrics did, whereas in the 3–8 Hz band, all three metrics showed significant differences. Violin plots of subject-level averages show non-overlapping $\mathcal{H}_A$ distributions in both bands (panels a and c), and subject-wise boxplots further illustrate ExSEnt's ability to distinguish gait from freezing (Fig. 9, panels b and d).

While SampEn provides significant results (Table 2), it only provides an aggregated index of complexity. ExSEnt, on

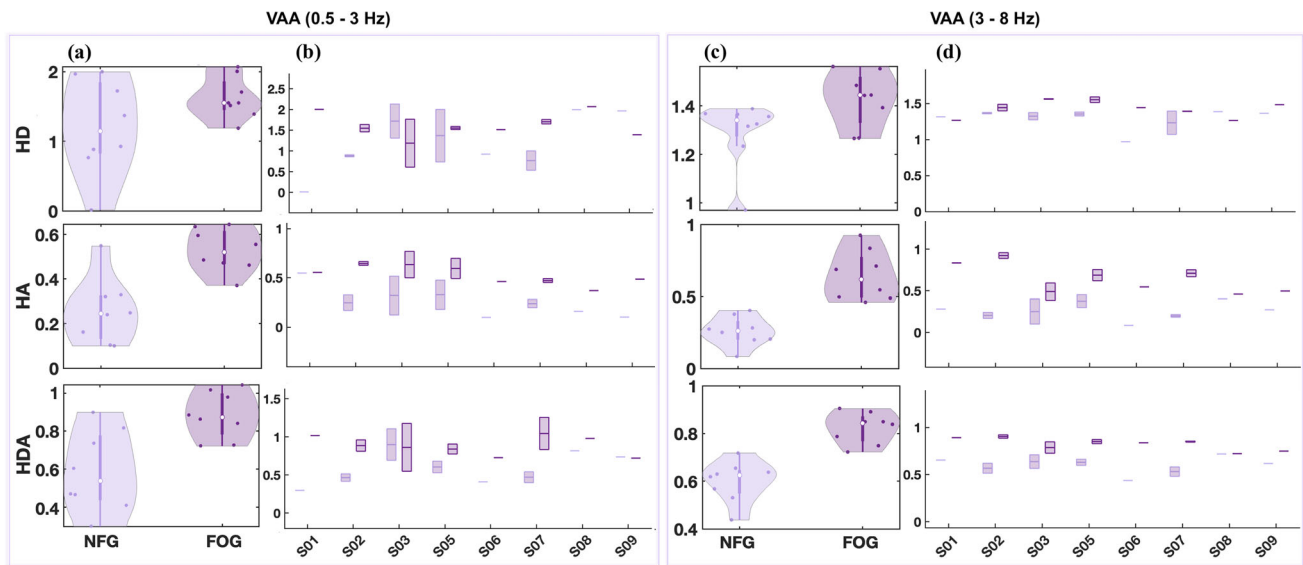


Fig. 9 ExSEnt metrics of vertical ankle acceleration (VAA) in Parkinson's disease. Panels (a) and (b) show the 0.5–3 Hz band, and c and d the 3–8 Hz band of the band-pass filtered VAA for all three metrics. Panels (a) and (c) show the violin plots comparing no-freezing gait (NFG)

and freezing of gait (FOG) using the subject-level mean of each metric. Panels (b) and (d) depict the subject-wise boxplots for NFG and FOG, for the ExSEnt metrics, showing per-subject distributions

the other hand, shows that the NFG-FOG difference is dominated by amplitude-related variability ($\mathcal{H}_A$, $\mathcal{H}_{DA}$ significant in both bands), while purely temporal variability is absent in 0.5–3 Hz and only marginal in 3–8 Hz. Thus, ExSEnt adds mechanistic interpretability and enables hypothesis-driven attribution of the observed complexity change, beyond the univariate separation provided by SampEn.

4 Discussion

The ExSEnt metrics provide deeper insight into the structural organization of nonlinear dynamical systems by distinguishing the components of entropy—whether it arises primarily from timing variability (segment durations) or from amplitude variability (segment-wise net changes). Across bifurcation diagrams, the three metrics, $\mathcal{H}_D$, $\mathcal{H}_A$, and the joint $\mathcal{H}_{DA}$, offer complementary perspectives on the underlying dynamics. Notably, these measures do not simply co-vary; instead, they differentiate temporal from amplitude-driven contributions to complexity, revealing structure that a single, undifferentiated complexity score could not unravel.

In the bifurcation diagrams (Figs. 3–5), we observed in different intervals and regimes, either $\mathcal{H}_D$ or $\mathcal{H}_A$ dominated their joint entropy and $\mathcal{H}_{DA}$ closely tracked the dominant component. This indicated that one feature (timing or amplitude) was the primary driver of system complexity in those parameter ranges. Such feature-specific dominance was not captured by traditional complexity measures that conflate

irregularities in timing and amplitude. By isolating the contributions of durations and amplitudes, ExSEnt yielded a more nuanced characterization of the trajectory, exposing patterns in the temporal organization or in the amplitude modulation that conventional entropy metrics overlooked.

There were also regions of the bifurcation diagrams (e.g., Fig. 5, $0.5 < \sigma < 0.7$) where the trajectories exhibited a high density of extrema while the ExSEnt indices remained comparatively low relative to other ranges (e.g., $-1.5 < \sigma < 0.4$). This dissociation suggested that frequent extrema alone did not imply high complexity; rather, the distribution of extrema followed a more regular pattern, reducing entropy. In that referred region, both $\mathcal{H}_D$ and $\mathcal{H}_A$ exceeded the joint entropy $\mathcal{H}_{DA}$, implying partial redundancy between temporal and amplitude variations (i.e., some predictability of one feature given the other).

On representative biological data, ExSEnt provided interpretable, feature-level insights. In the EMG recordings, the amplitude-driven entropy $\mathcal{H}_A$ decreased during movement execution relative to baseline, while the temporal component $\mathcal{H}_D$ remained approximately unchanged. Although EMG amplitudes increased when muscles were engaged, the pattern of amplitude changes at extrema became less complex than at rest; this was reflected proportionally in the joint index $\mathcal{H}_{DA}$. Thus, ExSEnt separated a power/gain increase from a reduction in amplitude complexity, which a single-value entropy would have conflated.

In VAA from participants with PD, ExSEnt cleanly separated band-specific effects of FOG. In the 3–8 Hz band, all

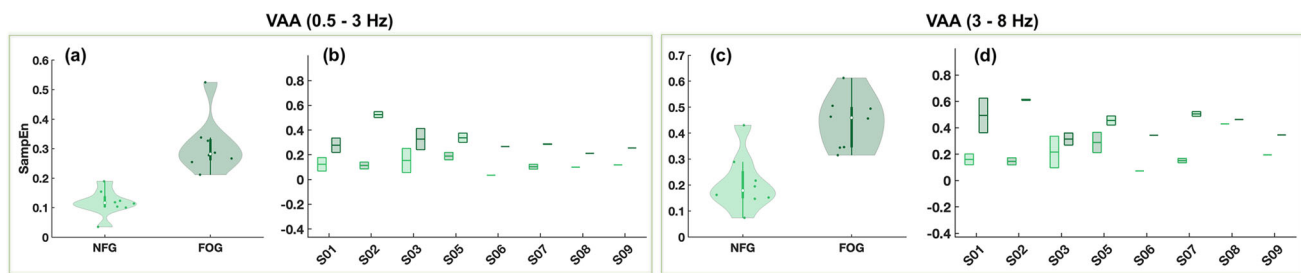


Fig. 10 Sample entropy of vertical ankle acceleration (VAA) in Parkinson's disease. Panels (a) and (b) show the 0.5–3 Hz band, and (c) and (d) the 3–8 Hz band of the band-pass filtered VAA. Panels (a) and (c) show the violin plots comparing the SampEn of no-freezing of gait (NFG)

and freezing of gait (FOG) using the subject-level mean of each metric. Panels (b) and (d) depict the subject-wise boxplots for NFG and FOG, showing per-subject distributions

three metrics ($\mathcal{H}_{\mathcal{D}}$, $\mathcal{H}_{\mathcal{A}}$, $\mathcal{H}_{\mathcal{DA}}$) increased, indicating concurrent rises in temporal and amplitude irregularities. In contrast, within 0.5–3 Hz only $\mathcal{H}_{\mathcal{A}}$ increased, implicating amplitude-driven changes at slow time scales while timing variability ($\mathcal{H}_{\mathcal{D}}$) remained stable. This accords with prior FI-based evidence that FOG is characterized by enhanced 3–8 Hz activity with comparatively stable locomotor band content [82, 83]. ExSEnt makes the attribution explicit by showing that, in the locomotor band, durations preserve their entropy while amplitudes exhibit increased complexity.

Furthermore, in a recent work, we have shown that ExSEnt metrics provide stable and informative biomarkers for dementia detection from EEG [84]. In this study, we extracted a set of classical spectral and complexity-based features, along with ExSEnt metrics, and demonstrated that timing ExSEnt metrics in the alpha and theta bands were the most reliable features for classifying healthy vs. dementia patients, supporting the ability of ExSEnt to discriminate between healthy and pathological neural dynamics.

ExSEnt defines each segment duration $\mathcal{D}_i$ as the integer number of samples between successive extrema, so the duration sequence $\{\mathcal{D}_i\}$ is inherently quantized by the sampling interval dt (i.e., durations lie on a dt -spaced grid). Consequently, if the sampling rate is too low to resolve the relevant extrema dynamics, distinct continuous-time durations may collapse to identical sample counts, reducing the apparent variability of temporal extrema patterns. Under adequate sampling (observed Nyquist criterion, where extrema dynamics are well resolved), changes in dt induce at most small perturbations of extrema locations, so the resulting quantization of $\{\mathcal{D}_i\}$ is expected to have a minor effect on $\mathcal{H}_{\mathcal{D}}$ relative to the dynamical variability of segment durations.

ExSEnt effectively characterized complexity variations in both canonical dynamical systems and real-world physiological signals. The joint entropy $\mathcal{H}_{\mathcal{DA}}$ provided a clear quantitative index that was consistent with established entropy-based assessments, while the separate components

identified the primary contributor to complexity (temporal vs. amplitude). In some cases (e.g., Fig. 6), the results indicated differing dimensionalities for temporal and amplitude-driven dynamics, reinforcing that these features need not vary on the same manifold. Peaks and drops in ExSEnt measures aligned with transitions between periodic and chaotic behavior, capturing critical dynamical shifts. Importantly, incorporating a noise-tolerance threshold via the tunable parameter λ improved robustness to small perturbations and observational noise, enhancing suitability for real, noisy systems.

In summary, ExSEnt provides a feature-driven perspective on signal complexity that complements traditional measures. By attributing overall entropy to timing and amplitude components and quantifying their interaction, ExSEnt improves interpretability and offers practical advantages for analyzing nonlinear dynamics in both synthetic and real data.

5 Conclusion

We introduced ExSEnt, a feature-decomposed complexity metric that characterizes the dynamics of nonlinear signals by separating temporal from amplitude contributions. The method segments a signal into monotonic portions defined by the sign of first-order increments, with an IQR-based threshold to provide robustness against noise-driven fluctuations. From each segment, we extract the duration $\mathcal{D}$ and the net amplitude change $\mathcal{A}$, then compute sample entropy on the duration sequence ($\mathcal{H}_{\mathcal{D}}$), the amplitude sequence ($\mathcal{H}_{\mathcal{A}}$), and their joint representation ($\mathcal{H}_{\mathcal{DA}}$). Using SampEn to quantify irregularities is advantageous in this context, since segmentation reduces the sample size—the number of extrema is typically smaller than the number of raw samples—and SampEn is known to perform more reliably on short sequences than other entropy measures, such as ApEn.

Comparing these metrics yields interpretable insight into where complexity originates. High $\mathcal{H}_{\mathcal{D}}$ indicates irregular timing of salient events, whereas high $\mathcal{H}_{\mathcal{A}}$ reflects variabil-

ity in amplitude excursions. The joint term $\mathcal{H}_{\mathcal{DA}}$ quantifies coupling: values lower than both individual components suggest redundancy (predictability of one feature from the other), while similarly high values across all three indicate independent temporal and amplitude contributions to overall complexity.

Although ExSEnt is presented here using SampEn as the underlying similarity-based estimator, the framework is not restricted to SampEn. Since ExSEnt operates by transforming the signal into extrema-segment descriptor sequences and then quantifying their pattern regularity, the same pipeline can be paired with alternative template-matching entropy estimators (e.g., FuzzyEn [85] or Multiscale entropy profiling [17]). In this work, SampEn was used as a widely adopted and computationally efficient baseline to isolate the contribution of the extrema-based temporal/amplitude decomposition; future work may explore other pattern estimators within the same framework.

Empirically, ExSEnt tracked complexity changes along control parameter sweeps in the Logistic map, Rössler system, and Rulkov map, aligning with transitions between periodic and chaotic behavior. In biosignals, ExSEnt isolated feature-level effects (e.g., band-specific changes in ankle acceleration during FOG vs. NFG, and amplitude-complexity reduction in EMG during movement despite power increases), providing a richer perspective than unidimensional entropy measures. The increment-based segmentation and noise-tolerance threshold improved robustness to small perturbations, making ExSEnt suitable for real-world, noisy recordings.

ExSEnt has broad applicability wherever disentangling timing and amplitude variability is informative, notably in biosignal analysis (e.g., EEG/MEG, EMG, ECG, and kinematics signals for applications ranging from sleep and anesthesia monitoring to seizure detection), as well as in finance, climate and environmental monitoring, and condition-based maintenance of engineered systems. Future work includes multiscale extensions (to attribute feature-specific complexity across time scales), principled parameter selection and theoretical bounds, multivariate/graph extensions for networked signals, the use of application-specific relevant events instead of extrema, and real-time implementations for embedded and clinical settings. Furthermore, ExSEnt metrics can be used together with state-of-the-art AI algorithms for time series classification.

In sum, ExSEnt complements traditional entropy methods by attributing complexity to distinct signal features and quantifying their interaction, thereby improving interpretability and advancing the analysis of complex dynamical systems and signals.

A Mathematical models

A.1 Stochastic signal generation

To validate the entropy-based metrics, we generated three canonical stochastic signals over $N = 2 \times 10^4$ samples with a sampling rate of $f_s = 200$ Hz. Each signal was simulated 100 times with fixed random seeds to ensure reproducibility. The statistical descriptors reported in Table 1 were computed across runs. Below, we summarize the mathematical formulation of these signals.

A.1.1 Gaussian white noise

Gaussian white noise was generated as samples from the standard normal distribution, $\mathcal{N}(0, 1)$, whose probability density function is

$$p(x) = \frac{1}{\sqrt{2\pi}\sigma^2} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right), \quad (16)$$

with mean $\mu = 0$ and standard deviation $\sigma = 1$, generated with the MATLAB command `randn(N, 1)`.

A.1.2 Pink noise

Pink noise ($1/f$ noise) was generated as a stochastic process with a power spectral density inversely proportional to frequency:

$$S(f) \propto \frac{1}{f}, \quad f > 0, \quad (17)$$

which yields equal energy per octave. In practice, it was generated using MATLAB's `dsp.ColoredNoise('Color', 'pink', N, 1)` function.

A.1.3 Brownian motion

Brownian motion was generated as the cumulative sum of independent Gaussian increments:

$$x[n] = \sum_{k=1}^n \eta[k], \quad \eta[k] \sim \mathcal{N}(0, 1). \quad (18)$$

This corresponds to the discrete-time equivalent of integrated white noise, with a power spectral density scaling as

$$S(f) \propto \frac{1}{f^2}. \quad (19)$$

In MATLAB, it was implemented as `cumsum(randn(N, 1))`.

A.2 Nonlinear dynamical models

We used three nonlinear systems (Sect. 3.2) to examine how the ExSEnt measures vary across a range of control parameters. Below, we briefly introduce these systems.

A.2.1 Logistic map

The logistic map is a simple nonlinear dynamical system defined by the recurrence relation:

$$x_{n+1} = rx_n(1 - x_n),$$

where $x_n \in [0, 1]$ represents the population at iteration n , and r is the control parameter governing the system's behavior. For $0 < r \leq 4$, the map exhibits a range of dynamical behaviors, including fixed points, periodic cycles, and chaos.

A.2.2 Rössler system

The Rössler system is a set of three coupled first-order autonomous differential equations that describe a continuous-time chaotic attractor:

$$\begin{aligned}\frac{dx}{dt} &= -y - z, \\ \frac{dy}{dt} &= x + ay, \\ \frac{dz}{dt} &= b + z(x - c),\end{aligned}$$

where a, b, c are system parameters that influence the system's behavior. In particular, c serves as a control parameter, impacting the transition between periodic and chaotic dynamics. A commonly studied set of parameters is $(a, b, c) = (0.2, 0.2, 5.7)$, which produces chaotic dynamics. The system was numerically integrated using the custom `ode4` solver, which implements the classical fourth-order Runge–Kutta method. This non-adaptive algorithm evaluates the derivatives four times per integration step and updates the state with a weighted average of these increments. The integration was performed with a fixed time step of $\Delta t = 0.1$, providing a sufficiently fine temporal resolution to capture the system's chaotic oscillations.

A.2.3 Rulkov map

The Rulkov map is a two-dimensional discrete-time system designed to model neuron-like bursting dynamics. It consists of a fast variable x_n , representing the membrane potential, and a slow variable y_n , which modulates the system's state

over time:

$$x_{n+1} = \frac{\alpha}{1 + x_n^2} + y_n,$$

$$y_{n+1} = y_n - \mu(x_n - \sigma).$$

The parameter α primarily controls the excitability of the system and typically ranges from 2 to 6, where lower values correspond to tonic spiking and higher values induce chaotic bursting. The small parameter μ , governing the slow evolution of y_n , is usually set within $10^{-4} \leq \mu \leq 10^{-2}$, ensuring a clear timescale separation between fast and slow dynamics. The parameter σ acts as an external input or bifurcation parameter, influencing transitions between different dynamical regimes, with common values in the range $-1 \leq \sigma \leq 1$. The interplay between these parameters allows the Rulkov map to generate a variety of neuronal activity patterns, including periodic spiking, bursting, and chaotic oscillations.

Author Contributions S.K. conceived the methodology, developed the codes, performed the analyses, and wrote the manuscript. F.B. and P.V. made substantial contributions through regular discussions, proposed the inclusion of real datasets and representative signals, guided the framing and interpretation of results, and critically revised the manuscript for intellectual content. F.B. and P.V. also ensured analytical rigor and clarity by conducting critical revisions of the manuscript and overseeing validation choices. All authors reviewed and approved the final version of the manuscript and agree to be accountable for all aspects of the work.

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Data Availability Two publicly available datasets are analyzed in this work: 1. Freeze of gait in healthy individuals vs. Parkinson's patients, using the Daphnet dataset: <https://archive.ics.uci.edu/dataset/245/daphnet+freezing+of+gait> 2. Finger pinching EMG, using the Motor Movement dataset: <https://gigadb.org/dataset/100295> (Cho H, Ahn M, Ahn S, et al. Supporting data for "EEG datasets for motor imagery brain computer interface." GigaScience Database, 2017). All codes are publicly available at: https://github.com/GNB-UAM/Kamali_et_al_ExSEnt_Extremasegmented_Entropy_Analysis_of_Time_Series-GNB-UAM/Kamali_et_al_ExSEnt. The ExSEnt implementation is also integrated into the EEGLAB plugin for entropy-based measures [86], accessible at: <https://github.com/amisepa/Escape>.

Declarations

Conflict of interest The authors declare no conflict of interest.

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